

From Biological Question to Mathematical Problem

Research Process Essay (January 2025 - Present)

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I am drawn to systems that resist clean explanations. When a process is both highly studied and still difficult to explain, I instinctively want to understand why. This curiosity has always pulled me toward problems where uncertainty is part of the challenge and there is no obvious path to the answer.

The 18-month college-level Program for Algorithmic and Combinatorial Thinking introduced me to abstraction and formal reasoning. Almost three years on my school's competitive physics team trained me to build quantitative models from first principles, often from little more than a physical system, a set of observations, and a blank page. This trajectory culminated when I served as lead theorist at our first-place finish in the U.S. Invitational Young Physicists Tournament. Presenting and defending our reasoning before a panel of expert physicists and peers reinforced a lesson that became central to my research. Convincing explanations require clarity, sound technical work, and strong communication under critique.

Throughout high school, I was most interested in math, physics, and biology, but I found biological systems to be the most intricate and unforgiving. I became especially drawn to the immune system because it seemed difficult to reduce to any single mechanism, which led me to design an independent study in statistical immunology during my junior year. As I explored immune dysregulation with my teacher, rheumatoid arthritis presented a puzzle that refused to leave me alone. Why has RA incidence in the United States continued to rise since 1990 when two of its best-established risk factors, smoking and genetic susceptibility, have either declined or remained relatively stable at the population level? Conversely, obesity rates had increased sharply over the same period, and I began to wonder whether rising adiposity could be contributing to RA pathogenesis rather than merely coinciding with it.

When junior year ended, the independent study no longer provided a regular structure, but I continued pursuing the question on my own throughout the summer. I initially tried regression models to surface patterns, only to discover that young-onset RA datasets were limited, and more fundamentally, that observational data could not directly reconstruct exposures that might precede diagnosis by decades or reveal the counterfactual population outcomes I wanted to study. I eventually connected with Michael Bryan, an MD/PhD student at Oxford, who became my research mentor. Through our discussions, I realized early on that waiting for the 'perfect dataset' was itself a methodological trap. That realization forced me to rethink the problem and pivot toward simulation. I knew simulation could not recover exposure histories that had never been observed or prove causality from missing information, but it could make my assumptions explicit and let me test counterfactual scenarios under those assumptions.

Eighteen months in, my research has evolved through two stages and is now entering a third. Stage I moved from a biological question to a computational model built around stochastic simulation of long-latency disease risk. Stage II emerged when I began treating the model itself as something to understand. I started asking harder questions about what its assumptions allowed me to infer and where those assumptions might break down. Now, with recently secured access to longitudinal patient data, I am preparing to test the framework against real observations. Stage III is still being formulated in my head, but I am most interested in whether its successes and failures reveal a deeper mathematical or statistical problem.

Stage I - From Biological Question to Computational Model

Once I committed to simulation, I began building the framework from the literature up. I built the simulation engine in Java as a Monte Carlo framework generating synthetic populations of one million individuals, large enough to keep sampling noise small relative to the effects I wanted to examine. Rather than assigning fixed outcomes, I modeled individual risk probabilistically using published incidence estimates, odds ratios, and demographic factors, enabling me to compute prevalence patterns across age groups. I implemented an age-dependent exponential decay function to model attenuation of BMI-related risk over time and systematically perturbed obesity distributions to examine counterfactual population scenarios.

As an external consistency check, I compared predicted RA prevalence against Global Burden of Disease benchmarks and later used age-specific targets in calibration of selected model parameters. I also performed sensitivity analyses on key assumptions, including the BMI decay constant, to see whether plausible changes materially altered the conclusions. Each scenario was evaluated across 2,500 **stochastic realizations**, while uncertain parameters and assumptions were varied separately rather than treating repeated simulations as estimates of parameter uncertainty. This distinction became increasingly important as the project developed. As I refined the model, the differences between stochastic variability, parameter uncertainty, and model discrepancy became much clearer. Increasing the simulated population or number of Monte Carlo realizations can reduce numerical uncertainty in estimated model outputs, but it cannot make uncertain parameters known or correct an inadequate model structure. That pushed me to think more carefully about the difference between uncertainty within the model and disagreement caused by the model itself.

Several early versions of the model produced implausible outputs and unstable results. As the implementation grew more complicated, the code also became increasingly difficult to debug. Because this work was not tied to a grade or deadline, I was able to experiment freely, and that freedom proved essential. Rather than continuing to patch individual errors, I rebuilt the simulation several times from the ground up, making the framework increasingly modular and forcing myself to reconsider the assumptions embedded within it. In retrospect, many of the most important advances in my work came from the model's failures because they exposed gaps in my own reasoning.

Through repeated revisions and discussions with Michael, I gradually constrained a plausible range for BMI's modeled population-level contribution to RA prevalence. Using Java for simulation and Python for analysis and visualization, I moved from rebuilding the framework to more targeted calibration and sensitivity analysis. Once the outputs stabilized within epidemiologically plausible bounds, the research began to feel less like constructing a model and more like understanding what the model could actually support.

Developing this model forced me to confront a broader methodological issue. I had built a simulator that could reproduce plausible population-level patterns, but I began to question what that agreement actually told me about the underlying system. Long-latency exposures are especially difficult because the relevant histories may be incomplete or entirely unobserved. Simulation cannot recover that missing information, and its conclusions remain conditional on the risk relationships, exposure distributions, and structural assumptions supplied to the model. What it can do is make those assumptions explicit and test their consequences through counterfactual scenarios. But even a simulator that reproduces an observed prevalence curve is not necessarily the correct model of the underlying process, because different parameter combinations, exposure histories, or model structures may produce similar observable behavior. That distinction began to shift my attention from whether the model worked to what the model could actually allow me to infer.

After several months of revising the model, I began drafting a manuscript and submitted an abstract to the American Association of Immunologists Midwinter Conference. After scientific review, I was accepted to present and was the conference's only high school student presenter. Preparing the work for a professional audience forced me to compress months of modeling decisions into a clear argument, which sharpened both my conclusions and my understanding of the framework's assumptions and limitations. At the conference, I presented the model to clinical researchers and fielded questions about its structure, interpretation, and potential beyond rheumatoid arthritis. The experience was transformative as it pushed me to think more carefully about which aspects of the work were specific to the biological application and which might reflect a broader methodological problem.

Stage II — From Computational Model to Inference Problem

The conference also clarified that the framework was ready to be tested against real patient data. I posted the manuscript as a first-author preprint and began reaching out to clinical researchers to request access to longitudinal cohorts. After numerous proposals, rejections, calls, and revisions, I received approval to work with de-identified patient data from Seattle Children's Hospital and the restricted ERAS-ERAN cohorts through King's College London. I ultimately centered the next stage of my research on ERAS-ERAN, where my research proposal was reviewed and approved by the cohort's Steering Group.

Securing that access has been pivotal. What I had originally planned as a final validation step has become an opportunity to test not only whether the model matches real observations, but where its assumptions, parameterization, and inferential structure hold up and where they begin to break down. Just as importantly, I've come to realize that disagreement with the data would not by itself reveal *why* the model might fail. It could reflect poorly estimated parameters, stochastic variation, measurement error, omitted structure, or broader model misspecification. Distinguishing among those possibilities is now the central question driving the next stage of my research.

After graduation, the summer again gave me the freedom to work without the constraints of a busy school schedule. As I prepared for what I thought would be a validation step, I began treating the framework less as a biological simulator and more as a mathematical object in its own right. I started asking what the model was actually doing as an inference system, which quantities were observed, which remained latent, and how assumptions about exposure histories and risk relationships entered the model.

Framing the model this way changed the questions I was asking. Instead of only asking whether the simulator could reproduce plausible population-level behavior, I began asking what information about the underlying states and parameters could actually be recovered from partial longitudinal observations, and under what assumptions that recovery would be identifiable and stable. I also began asking whether some quantities might be fundamentally non-identifiable. If distinct latent histories, parameter combinations, or model structures produce essentially the same observations, additional computation alone cannot determine which generated the real system.

As that shift became more concrete, I separated the new work from the original simulation codebase and created a second repository, *longitudinal-risk-inference*, focused on empirical validation, inference, model diagnostics, and uncertainty quantification. I kept the original repository intact as a record of the completed simulator, while the new one became the workspace for testing the framework against longitudinal observations and developing the mathematical tools needed to understand where it succeeds and fails.

This new direction has pushed me toward questions of uncertainty, model discrepancy, and robustness. I am especially interested in separating uncertainty about unknown parameters from structural error in the model itself, since even a precisely estimated parameter would mean little if the underlying model were wrong. I am also thinking more carefully about which parts of the framework should remain mechanistic and which, once the approved data are transferred for analysis, might instead be estimated or learned from data. My central question has now changed. I am no longer only asking whether the simulation can reproduce a plausible biological pattern. I want to know what can actually be inferred from partial observations of a stochastic system, under what assumptions that inference is identifiable and stable, and where the framework's limitations might reveal a more general mathematical or statistical problem.

Stage III - Empirical Validation and Methodological Extension

The next stage of my research is therefore not simply to validate the existing framework, but to use real longitudinal observations to find where its assumptions or structure begin to break down. Comparing the simulator against ERAS-ERAN should help me determine whether discrepancies arise from uncertain parameters, stochastic variation, weakly identifiable latent structure, model misspecification, or relationships that would be better estimated from data than imposed mechanistically.

These possibilities are related, but they imply very different problems. If disagreement with the data can be explained by uncertainty in the parameters or stochastic variation, the existing model may still be adequate. If it instead reflects missing structure or systematic model discrepancy, then improving parameter estimates will not solve the problem. And if the available observations cannot distinguish between different latent histories or parameter values in the first place, then the limitation is one of identifiability rather than computation. Learning to tell these possibilities apart is now one of the central challenges of the project.

I do not yet know which of these limitations will prove most important, and I do not want to choose a mathematical method before the data reveal what problem actually needs to be solved. My first task is therefore to determine what the longitudinal observations can actually tell me about the model. That might expose an identifiability problem, reveal systematic model discrepancy, or show that certain mechanistic relationships would be better learned from data. It may also raise questions I have not anticipated. I want the mathematical direction of this stage to emerge from the limitations exposed by the data rather than forcing the data into a problem I have already decided to solve.

Even if the model fits the longitudinal data well, that alone will not establish that it has recovered the true underlying mechanism. Different latent processes may still produce similar observations, and some assumptions may remain impossible to test with the available data. I therefore want to understand not only whether the model fits, but what aspects of the underlying system the data actually constrain and which conclusions remain dependent on assumptions that the observations cannot resolve.

What matters to me is not making the framework mathematically more elaborate for its own sake, but identifying a limitation whose resolution requires a genuinely new method or result. The biological application gives me a concrete system in which those limitations can be exposed and tested, but I increasingly see rheumatoid arthritis as a testbed rather than the endpoint of the research. If the same inferential difficulty arises from partial observation, latent structure, model misspecification, or the interaction between mechanistic and learned components, then the question may extend well beyond this particular disease. My goal is to determine whether the model's successes and failures reveal such a problem and, if they do, formulate it precisely enough to study in its own right.

Looking Back to See Forward

Looking back, the project has changed how I approach difficult research problems. Failed models, unstable assumptions, and barriers to data access have often been useful because they showed me where an approach was inadequate or forced clearer thinking. Several of the questions I am now pursuing emerged because an earlier approach exposed a limitation I had not anticipated. As I enter college, I want to develop the mathematical tools needed to pursue questions like these more rigorously.